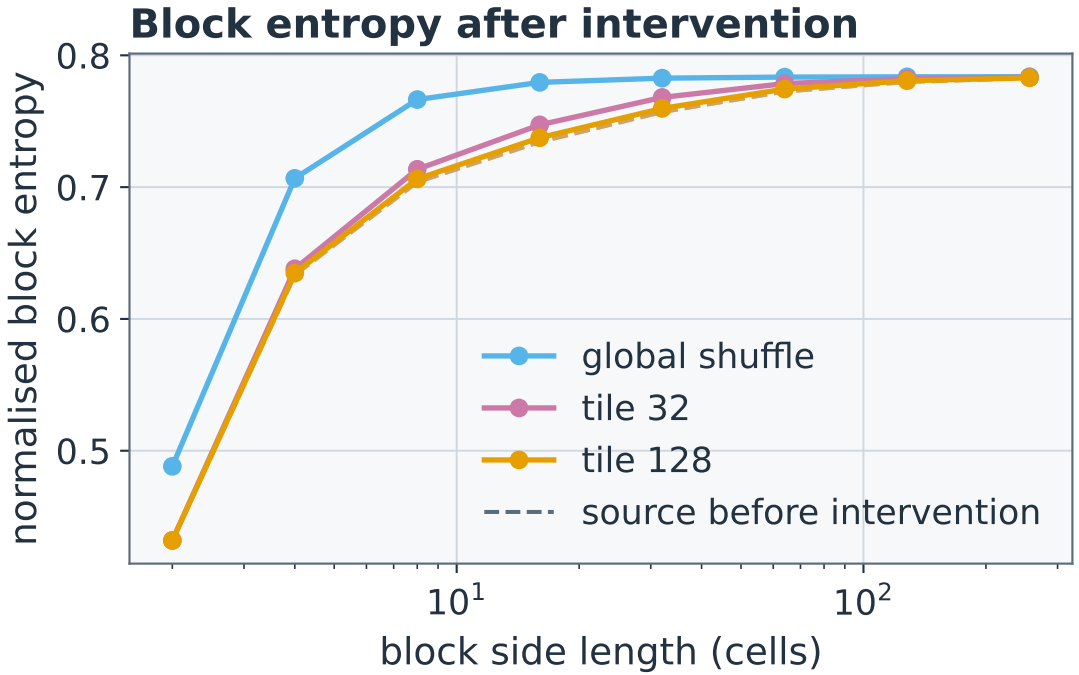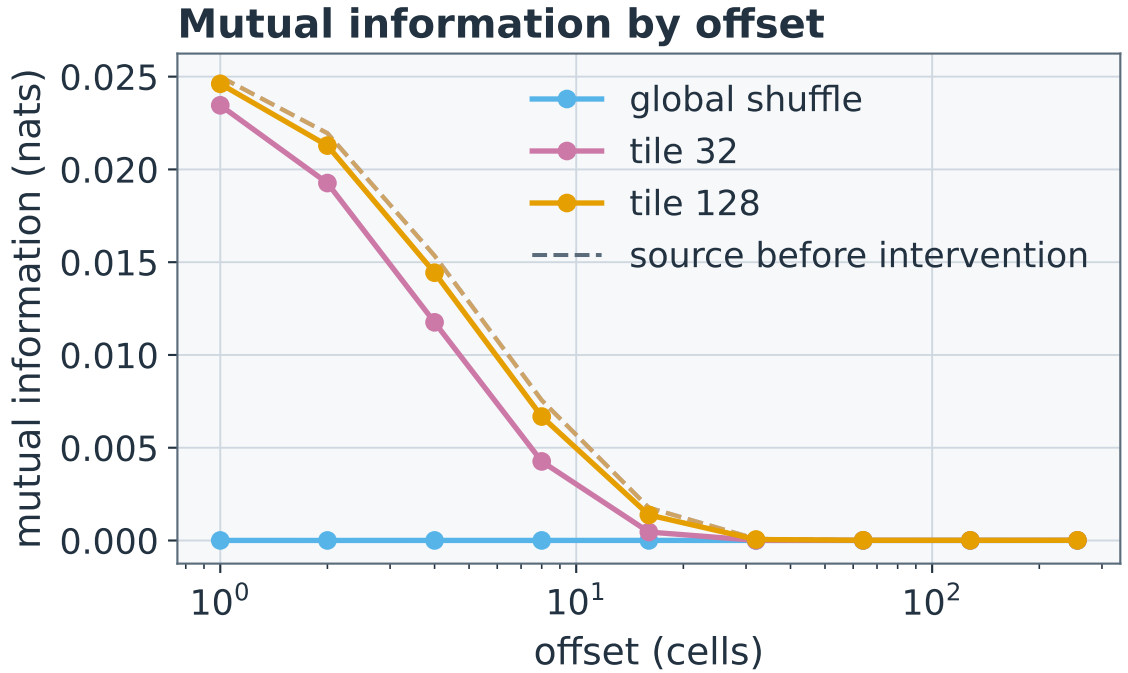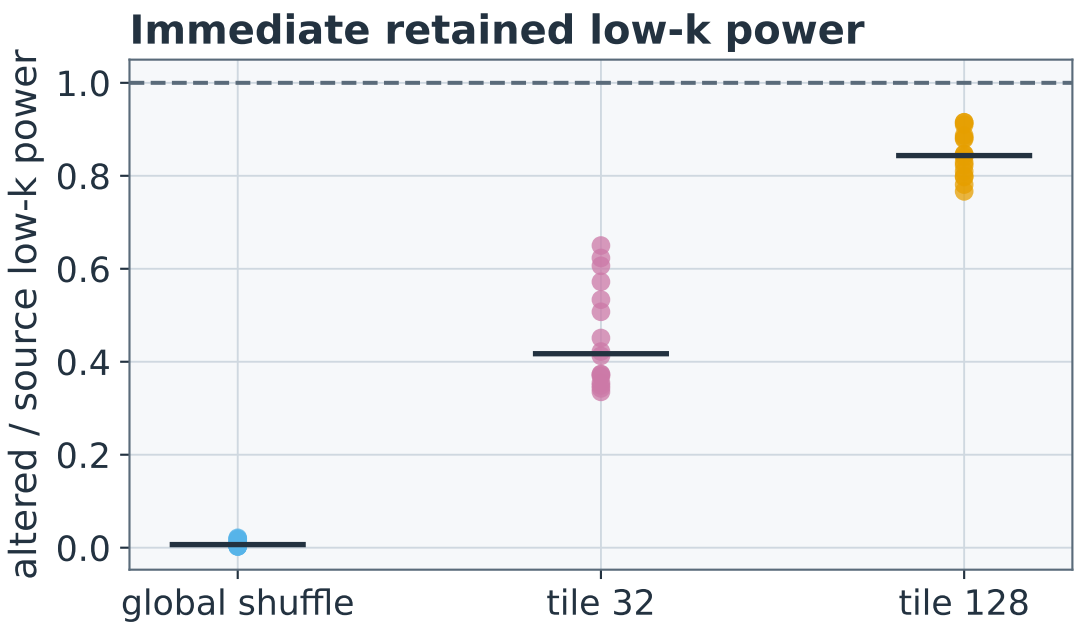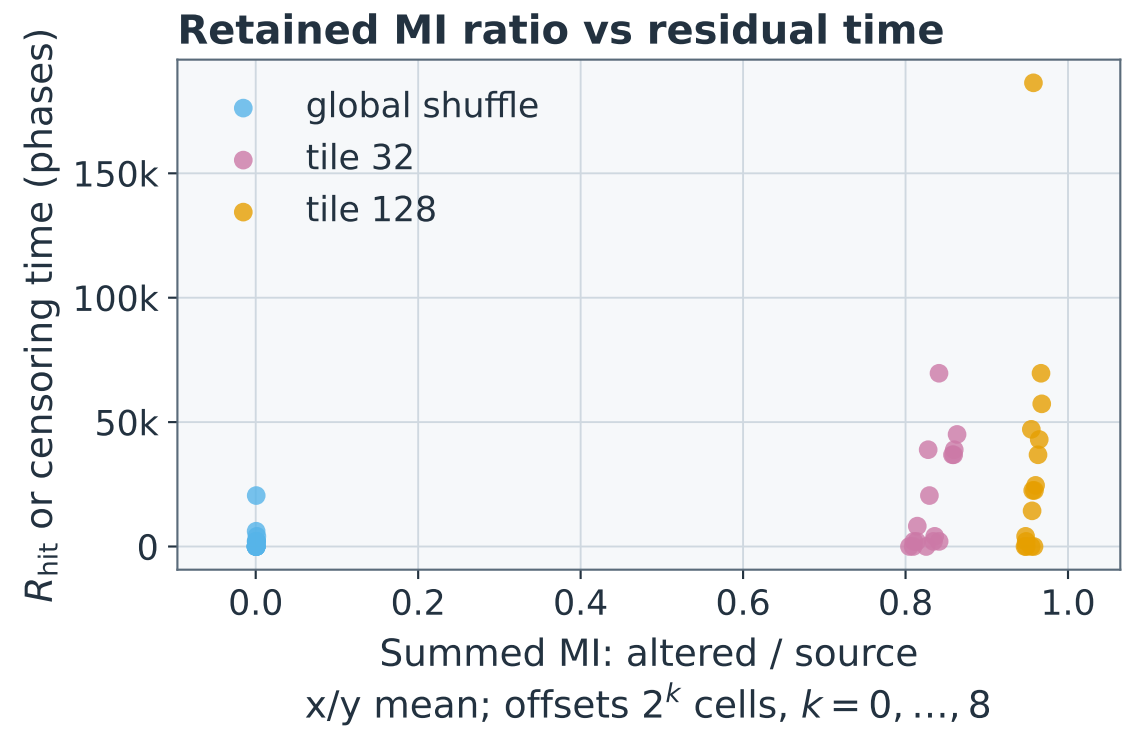How much spatial information survives each intervention?